Numerical Setup

S. Brennan
Max Planck Institute for Astrophysics, Garching, Germany

Numerical Setup

We model the double-explosion scenario with a two-stage pipeline coupling the multi-group radiation hydrodynamics code STELLA (Blinnikov et al., 1998, 2006) with the non-LTE radiative transfer code CMFGEN (Dessart et al., 2015; Hillier & Miller, 1998). In the first stage (SN_A), a Type IIb progenitor — the solar-metallicity $12\,M_\odot$ model `s12.0` of Woosley & Heger (2007), stripped to a $\sim 3\,M_\odot$ helium core and matched to model 12C of Ergon et al. (2022) for SN 2011dh — is exploded with a thermal bomb of 0.9×10^{51} erg deposited over 10 s in the inner $0.3\,M_\odot$, yielding a final kinetic energy of $\sim$0.7 foe. A thin hydrogen envelope of 0.03–0.10 $M_\odot$ ($X = 0.7$) is added to the surface, and a low-density ($\dot{M} = 10^{-6}\,M_\odot\,\mathrm{yr}^{-1}$) sigmoid-blended circumstellar wind extends to $\sim 10^{17}$ cm for numerical stability. A ^{56}Ni mass of $0.075\,M_\odot$ (Ergon et al., 2022) is distributed over the inner $0.3\,M_\odot$. STELLA evolves the explosion with 500 Lagrangian zones using a GEAR integrator ($\epsilon = 3 \times 10^{-3}$), with $E_{\mathrm{KO}} = 0$ to preserve the pre-SN velocity structure, `FitTau` = 5 for frequency-dependent opacity, and a minimum of $N_{\mathrm{zon}}/10$ diffusion zones to prevent late-time N_{cnd} collapse.

In the second stage (SN_B), the SN_A ejecta is homologously evolved to $\sim$400 d and serves as the CSM. A H/He shell ($1.0\,M_\odot$, $X = 0.7$, $Y = 0.3$) with a power-law density profile ($\rho \propto r^{-8}$) following Dessart et al. (2016) is placed at the centre and launched at 5000 km s^{-1} into the CSM. STELLA evolves the shell–CSM interaction for 45 d with $E_{\mathrm{KO}} = 0$ (preserving both shell and CSM velocities), enhanced artificial viscosity ($A_Q = 10$) for the strong velocity reversal, and no ^{56}Ni (the luminosity is entirely interaction-powered).

At selected epochs (30 d post-SBO for SN_A; 20–40 d post-launch for SN_B), snapshots are extracted from the STELLA output and remapped onto CMFGEN depth grids ($N_D \sim 150$) using an adaptive algorithm that resolves ionisation fronts and velocity reversals. CMFGEN computes non-LTE level populations and emergent spectra with `VEL_LAW = 12` (reading the non-homologous velocity directly), a tridiagonal depth-coupling solver for the non-monotonic velocity, Ng acceleration disabled to avoid trace-species instability at ionisation fronts, and temperature fixed to the hydrodynamic values. The inner boundary uses a diffusion condition seeded by the STELLA luminosity. The complete pipeline — progenitor conversion, STELLA evolution,

snapshot extraction, CMFGEN spectral synthesis, and dependency-chained job submission — is orchestrated by a single workflow manager on the MPCDF Viper HPC cluster.

References

Blinnikov, S. I., Eastman, R., Bartunov, O. S., Popolitov, V. A., & Woosley, S. E. 1998, ApJ, 496, 454, doi: 10.1086/305375

Blinnikov, S. I., Röpke, F. K., Sorokina, E. I., et al. 2006, A&A, 453, 229, doi: 10.1051/0004-6361:20054394

Dessart, L., Audit, E., & Hillier, D. J. 2016, MNRAS, 458, 2094, doi: 10.1093/mnras/stw440

Dessart, L., Hillier, D. J., & Audit, E. 2015, A&A, 579, A46, doi: 10.1051/0004-6361/201425403

Ergon, M., Fransson, C., Jerkstrand, A., et al. 2022, A&A, 666, A51, doi: 10.1051/0004-6361/202243565

Hillier, D. J., & Miller, D. L. 1998, ApJ, 496, 407, doi: 10.1086/305358

Woosley, S. E., & Heger, A. 2007, Physics Reports, 442, 269, doi: 10.1016/j.physrep.2007.01.004